# FLOODSHADOW Rainfall Intelligence and Evidence Verification for Guayaquil

**3Jota**

José Luis Paladines Sánchez

John Maximiliano Carreño Robalino

Julinho Jesús Plua Mendosa

Hack2Impact 2026  Ecuador challenge

**Abstract—FLOODSHADOW connects weather forecasts, territorial susceptibility, documented impacts and verification for intense rainfall in Guayaquil. The implemented prototype preserves 6652 official census polygons and groups them into 100 named analytical areas using a municipal nomenclature dataset and parish fallbacks. Two documented October 2026 storms and a separately preserved prospective weather capture support a connected before, rainfall replay and impact workflow. Observed station totals, model estimates, forecast outputs and candidate incident coordinates remain distinct. An explained index identifies places where evidence collection may be useful; it is not a calibrated flood probability. A separate Bayesian laboratory studies imperfect reporting and adaptive verification. Structural validation includes 38 passing automated tests, geometry checks and an interactive named-area filter. Real-event predictive effectiveness is not demonstrated: none of six calculable candidate sectors in the October 24–25 collection exceeds the exploratory priority threshold. The contribution is an auditable decision-support workflow and a transparent path toward field validation, rather than a claim of operational flood prediction.**

**Index Terms—rainfall, geographic information systems, underreporting, evidence provenance, active verification.**

## I  Problem and challenge alignment

The Ecuador challenge asks how intelligent solutions can help Guayaquil anticipate and respond better to intense rain [1]. Evidence is fragmented across weather products, territorial layers and incident lists. A report at an intersection does not establish that its entire neighborhood flooded. Equally, no report does not establish that a place was unaffected. Conflating these situations can make an attractive map misleading.

Our proposed users are analysts, researchers and staff reviewing where to collect evidence or investigate a rainfall episode. We support maps, data visualization, explainable prioritization and verification. The prototype does not issue official alerts, coordinate rescue operations, recommend safe routes or estimate economic losses. These functions require evidence and operational authority that the present dataset does not provide.

## II  Implemented solution

An event identifier connects expected rainfall, hourly replay, later records and accepted verifications. The English interface uses Storm summary, Before rain, Rainfall replay, Reported impacts and Research lab. A persistent area search isolates both map geometry and analytical denominators. A selection remains available when changing stages.

Users can switch between original census polygons and named-area outlines. Urdesa, for example, groups 89 complete polygons. Searching San Borondón resolves the Samborondón alias and shows its 246 INEC polygons separately. No Guayaquil forecast, priority or impact estimate is copied into that neighboring canton [2], [3].

| View | Question answered |
|---|---|
| Storm summary | What evidence exists for this storm? |
| Before rain | What rainfall did the earlier model run expect? |
| Rainfall replay | How did hourly model estimates change? |

| View | Question answered |
|---|---|
| Reported impacts | What locations were reported and what remains unverified? |
| Research lab | How can imperfect reporting and verification affect inference? |

Color legends explain meaning and units. Amber indicates relative watch priority. Blue indicates rainfall in mm or mm/h; its visible numerical scale follows the selected window. Red dots are provisional locations of published records. Gray means a missing value or lack of whole-sector verification, depending on the selected layer. Positive and negative sector-wide checks have separate states. The impact view starts with rainfall and report points rather than an entirely unverified gray surface.

## III Data and geographic integrity

INEC 2022 supplies 6652 unique canton-sector identifiers: 6416 under main-parish code 090150 and 236 rural polygons [2]. The main-parish code includes peripheral land and is not asserted to be a legal urban limit. Municipal nomenclature supplies 95 source polygons and recognizable place names [4]. A census polygon is assigned only when one named source area covers more than half of its original surface; remaining units use parish or peripheral groups. The resulting 100 groups form a complete, nonduplicated partition of the analyzed census units. Their outlines are analytical group boundaries, not a new cadastral delineation.

Geographic calculations use WGS84 geodesic area and GEOS intersections. The original source had 7297 overlap pairs larger than 1 m². A declared deterministic partition removes analytical overlaps while preserving source files. Changes over 1% of original sector area exclude the territorial index in 897 sectors. Twelve invalid simplifications fall back to valid analytical geometry. All 6652 rendered sector geometries and 737 technical zone aggregates are valid. Membership, method, source URL and source checksum are retained.

The documented susceptibility dataset contains 5746 polygons [5]. Their union is intersected with each analytical sector to prevent double counting. The effective date is unknown, so the footprint cannot prove that a location flooded in a particular past event. The interface describes it as susceptible land, not an incident history.

Weather products use six distinct sampled ECMWF IFS cells per episode, with nominal resolution about 9 km [6]. A cell within 12 km of a representative sector point can be assigned; this does not increase spatial precision. Rain totals require complete, unique hourly timestamps. Missing values remain missing. ESPOL published five differing station totals for October 19, ranging from 112 to 181 mm [7]; exact instrument coordinates and accumulation windows are unavailable, so we neither interpolate these totals nor calculate a station-based forecast error.

For October 24–25, the attributed Ecuavisa list contains 68 urban water-accumulation locations and ten other typed records, including three reported floods in Tenguel [8]. Eleven intersections have candidate coordinates. Only seven have geocoder alternatives consistent with one original sector; four remain sector-ambiguous. The remaining 67 records have no assigned coordinates. Point records never fill an entire polygon as confirmed flooding. Direct municipal infrastructure queries returned HTTP 500; drainage capacity, maintenance, obstructions, pumping and tides remain unknown.

## IV Analysis and verification method

The real-event vigilance index is a multicriteria convention. Let h be the fraction of sector land intersecting the susceptibility union. Let r be the midrank percentile of forecast accumulation among unique sampled cells: r = (Nless + 0.5 Nequal) / N. Dry forecasts contribute zero rain points. The index is

**I = 50h + 50r.** **(1)**

Both inputs must exist; missing factors are not silently reweighted. Equal importance and the threshold $I \geq 60$ are explicit choices, not parameters fitted to the listed incidents. No calibrated flood probability or statistical confidence interval is claimed. Shadow Detection selects high-index sectors without sufficient localized evidence for further investigation.

The original research laboratory separately models impact occurrence and reporting conditional on impact, with spatial latent fields and Poisson reporting, including false reports. Its design follows the general need to separate a hidden state from imperfect observation [9], but it is not a validated transfer of an ecological model to Guayaquil. Positive or negative experimental inspections reweight shared latent hypotheses and update other sectors. Priors and synthetic experiments remain explicitly experimental.

Actual verification requires a storm, official sector, observation time, observer, method, descriptive evidence, HTTP(S) source and scope. Point observations do not count as whole-sector inspections. A negative sector-wide result needs representative evidence for the entire completed storm window. Entries start pending, can be accepted by the local operator and may be corrected without erasing history. New accepted evidence changes states, metrics and the verification queue; it does not rewrite the original assessment or automatically train a useful predictor from a few examples.

Historical forecast runs were initialized before the selected storms, but the platform retrieved them later; reconstructed index values are not counted as predictions issued beforehand. A separate capture was received on October 25 at 18:34 UTC−5 for the window October 25 at 19:00 through October 28 at 19:00. Its inputs, index and SHA-256 checksum are preserved before that window. The provider’s exact issue time is unavailable. A checksum proves content consistency, not independently certified time or authenticity.

## V Validation and observed results

| Check or metric | Observed result |
|---|---|
| Automated tests | 38 passed; no failures |
| Census membership | 6652 units, one group each |
| Urdesa filter | 89 polygons; 3 mapped point records |
| October 24–25 comparison | 0 of 6 calculable candidate sectors at $I \geq 60$; 1 missing index |
| Real sector-wide verification | 0 accepted; affected share not evaluable |
| SQL integrity and foreign keys | Passed |
| Production build | Passed; MapLibre size warning remains |

The city’s model-assignment coverage is 99.44% of main-parish sectors but only 48.77% of their summed area. These denominators differ and neither is instrumental observation coverage. In the October 24–25 collection, seven sectors contain provisionally mapped points; this is not an estimate of the actual affected share of Guayaquil.

The unfavorable 0/6 exploratory index comparison is reported without changing weights or threshold. It is not forecast recall: the index was reconstructed later, point locations remain provisional and the report collection is incomplete. Unreported units are never used as negative ground truth. Real-event ranking superiority and forecast calibration remain unproven.

The preserved laboratory benchmark compared random selection, report counts, historical susceptibility, the original heuristic and the Bayesian model under the same candidate pools and inspection budget. The existing suite contains 2500 synthetic episodes. For hidden discoveries in its uneven-reporting setting, mean discoveries per five inspections were 0.980 for random/report-count selection, 1.632 for history, 1.636 for the original heuristic and 1.598 for the Bayesian model. Thus the model does not establish superiority over history in that setting. These are prior reproducible experimental results, not newly observed field outcomes; manifests and paired uncertainty estimates remain in docs/INFORME-v03.md.

## VI Expected impact and implementation plan

The intended impact is better evidence retrieval, clearer recognition of information gaps and a traceable basis for deciding what to inspect next. We have not measured reduced losses, faster emergency response or lives saved. Evaluation should measure time to find a source, correct interpretation of colors, coverage of representative inspections and discoveries with a demonstrably complete initial report collection.

| Phase | Next acceptance condition |
|---|---|
| Data stewardship | Confirm reuse terms, data owners and update cadence |
| Field protocol | Obtain representative positive and negative checks with compatible time windows |
| Prospective evaluation | Preserve forecasts before independent storms and compare fixed baselines |
| Model development | Fit only on training storms; assess calibration and held-out performance |
| Deployment | Shared database, authenticated review, monitoring and accessible usability tests |

The present application runs with React, Vite, MapLibre GL JS and local JSON snapshots. Python, Shapely/GEOS, pyproj and pyshp prepare geographic data; SQLite provides a portable query copy. Verification records currently use browser storage and JSON export. Development used AI-assisted coding with Codex; no generative model fabricates observations, forecasts or incident classifications. Members must understand and defend the implementation. External sources retain attribution; redistribution terms must be checked for each complete source dataset.

## VII Conclusion

FLOODSHADOW delivers a connected and inspectable rainfall-evidence workflow with named geographic filtering and explicit uncertainty. It answers the Guayaquil challenge through maps, explainable prioritization and verification. Its current defensible result is functional integration with reproducible structural checks. The next scientific milestone is representative field evidence and independent prospective evaluation, not a stronger claim based on the present limited labels.

## References


[1] IEEE-HKN Nu Zeta and Nu Eta, “Reglas del Hackathon Hack2Impact 2026,” competition brief supplied by the team, 2026.

[2] Instituto Nacional de Estadística y Censos, “Marco Geoestadístico 2022,” FeatureServer layer 7. Accessed: Oct. 25, 2026. [Online]. Available: https://idgn.ecuadorencifras.gob.ec/server/rest/services/Hosted/Marco_Geoestadistico_2022/FeatureServer/7

[3] Prefectura del Guayas, “Samborondón.” Accessed: Oct. 25, 2026. [Online]. Available: https://guayas.gob.ec/cantones-2/samborondon/

[4] Municipalidad de Guayaquil, “Sectores de nomenclatura,” Sistema de Información Local, updated Nov. 28, 2025. Accessed: Oct. 25, 2026. [Online]. Available: https://sil.guayaquil.gob.ec/territorial/sectores-de-nomenclatura/

[5] “Zonas Inundables GYE,” ArcGIS FeatureServer, layer 0, effective date not supplied. Accessed: Oct. 25, 2026. [Online]. Available: https://services7.arcgis.com/NWWHhu45fOJtCgG3/arcgis/rest/services/Zonas_Inundables_GYE/FeatureServer/0

[6] Open-Meteo, “Single Runs API,” technical documentation. Accessed: Oct. 25, 2026. [Online]. Available: https://open-meteo.com/en/docs/single-runs-api

[7] FIMCM, ESPOL, “Análisis meteorológico y oceanográfico El impacto de las inusuales precipitaciones de octubre de 2026 en Guayaquil,” 2026. [Online]. Available: https://www.fimcm.espol.edu.ec/es/noticias/analisis-meteorologico-y-oceanografico-el-impacto-de-las-inusuales-precipitaciones-de

[8] Ecuavisa, “68 sectores afectados por lluvias en Guayaquil,” Oct. 25, 2026. [Online]. Available: https://www.ecuavisa.com/amp/guayaquil/68-sectores-afectados-lluvias-guayaquil-tormenta-electrica-20261025-0011.html

[9] D. I. MacKenzie et al., “Estimating site occupancy rates when detection probabilities are less than one,” Ecology, vol. 83, no. 8, pp. 2248–2255, 2002, doi: 10.1890/0012-9658(2002)083[2248:ESORWD]2.0.CO;2.